HJ - KKT / Optimality Conditions Team TOB

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$$\min z = 3y_1 + 4y_2 + 5y_3$$

$$s.t.\ y_1 + y_2 \geq 1,\ y_2 + y_3 \geq 1,\ y_1 + y_3 \geq 1$$

$$0 \leq y_1, y_2, y_3 \leq 1.$$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix},\ b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\ C = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

Dual: $\max \pi_1 + \pi_2 + \pi_3$ · S.t. $\pi_1 + \pi_2 \leq 3$, $\pi_1 + \pi_2 \leq 4$, $\pi_2 + \pi_3 \leq 5$; $\pi_1, \pi_2, \pi_3 \geq 0$

red. c. $d_j = c_j - (A^T \pi)_j$;

Primal point: $y = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

a) Feasibility: Primal $\frac{1}{2} + \frac{1}{2} = 1$, $\frac{1}{2} + \frac{1}{2} = 1$, $\frac{1}{2} + \frac{1}{2} = 1$. W R: ~~ato~~ (c) 0

b) Dual Feas: $\hat{\pi} = (1, 3, 2)$ $1 \geq 0$ $3 \geq 0$ $2 \geq 0$ W R: (b) 0

c) $\hat{\pi}_i (A_i y - b_i)$

	π_i	Slack	Pivot slack
π_1	1	0	0
π_2	3	0	0
π_3	2	0	0

R: (c) 0

d) $d_1 = 3 - (\hat{\pi}_1 + \hat{\pi}_3) = 3 - (1 + 2) = 0$

$d_2 = 4 - (\hat{\pi}_1 + \hat{\pi}_2) = 4 - (1 + 3) = 0 \Rightarrow (\hat{d}) = (0, 0, 6)$

$d_3 = 5 - (\hat{\pi}_2 + \hat{\pi}_3) = 5 - (3 + 2) = 0$ (d) 0

e) $z_D = c^T y$ $z_D = 3\left(\frac{1}{2}\right) + 4\left(\frac{1}{2}\right) + 5\left(\frac{1}{2}\right) \Rightarrow z_D = 1{,}5 + 2 + 2{,}5 = 6$

$|c^T y - b^T \hat{\pi}| = |6 - 6| = 0$ R: (e) 0

All five optimality conditions are satisfied. Therefore, the primal point is optimal : $z_P = 6$.

b) $\pi' = (2, 2, 1)$ $y = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

$2 \geq 0$, $2 \geq 0$, $1 \geq 0$ (b) O.

All three primal constraints have zero slack

(c) O.

$d_1' = 3 - (\pi_1' + \pi_3') = 3 - (2+1) = 0$

$d_2' = 4 - (\pi_1' + \pi_2') = 4 - (2+2) = 0$

$d_3' = 5 - (\pi_2' + \pi_3') = 5 - (2+1) = 2$.

$d' = (0, 0, 2)$ $\Rightarrow$ (d) = 2.

(e) $c^T y = 6$

$b^T \pi' = 2+2+1 = 5$

Since $d' = (0+0, 2) \Rightarrow \sum \min(0, d_j') = 0$.

$|6 - (5+0)| = 1$. R: (e) 1

c) $\pi' = (2, 2, 1) \geq 1$. $\pi_1' + \pi_3' = 2+1 = 3 \leq 3$

$\pi_1' + \pi_2' = 2+2 = 4 \leq 4$

$\pi_2' + \pi_3' = 2+1 = 3 \leq 5$.

$c^T y \geq (A^T \pi')^T y \geq b^T \pi' \Rightarrow b^T \pi' = 2+2+1 = 5 \Rightarrow b^T \pi' = 5 \leq z_D = 6$.

$6 - 5 = 1 \rightarrow$ Distance = 1.

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